

Ejada Summer Internship 2026

Cloud Build Track – OCI & Terraform

Week 1 Deployment Report

OCI Foundations & Terraform Basics

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GitHub Repository: [ejada-intern-w1-terraform](#)

Contents

1	Introduction	2
1.1	Objective	2
2	Architecture Overview	3
2.1	Architecture Diagram	3
2.2	Component Summary	4
3	Network Setup	5
3.1	VCN Configuration	5
3.2	Internet Gateway	5
3.3	Public Subnet Configuration	6
3.3.1	Route Table	6
3.4	Security List	7
3.4.1	Ingress Rules	7
3.4.2	Egress Rules	9
4	Compute Instance Deployment	12
4.1	Instance Details	12
4.2	SSH Key Setup	16
4.3	Connecting to the Instance	16
5	File Storage Service (NFS) Setup	17
5.1	File System and Mount Target Creation	17
5.2	Mounting on the Instance	18
6	Block Volume Setup	20
6.1	Block Volume Creation	20
6.2	Attaching via iSCSI	20
6.3	Formatting and Mounting	21
7	Terraform Automation (Lab 2)	23
7.1	Overview	23
7.2	Repository Structure	23
7.3	Terraform Workflow	23
7.4	Terraform Outputs	24
7.5	File Storage NFS Mount via Terraform Environment	24

1. Introduction

1.1 Objective

The objective of this lab is to build a foundational cloud environment on OCI, first manually using the OCI Console (Lab 1), then reproduced entirely through IaC using Terraform (Lab 2).

The specific technical goals are:

- Deploy a Virtual Cloud Network (VCN) with a public subnet and internet connectivity.
- Launch a Linux compute instance (Oracle Linux 8, ARM-based) within that subnet.
- Attach a Block Volume to the instance for persistent disk storage.
- Provision a File Storage Service (FSS) file system and mount it on the instance via NFS.
- Codify the entire environment in Terraform and push to GitHub for version control.

2. Architecture Overview

2.1 Architecture Diagram

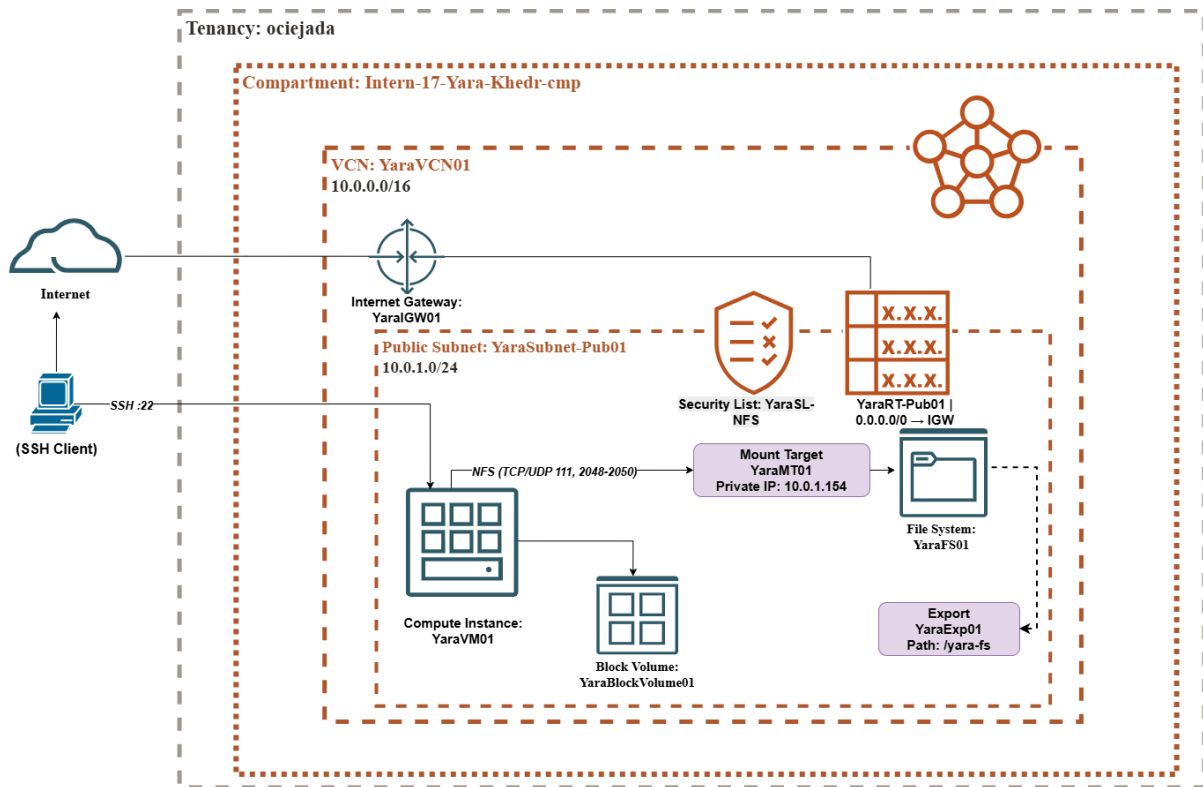


Figure 2.1: Week 1 – OCI Infrastructure Architecture Diagram

The diagram illustrates the full infrastructure deployed for Week 1. The environment resides within the `ociejada` tenancy, scoped to the `intern-17-yara-khedr-cmp` compartment. A Virtual Cloud Network provides the networking boundary, with a public subnet hosting the compute instance, block volume attachment, and file storage mount target.

2.2 Component Summary

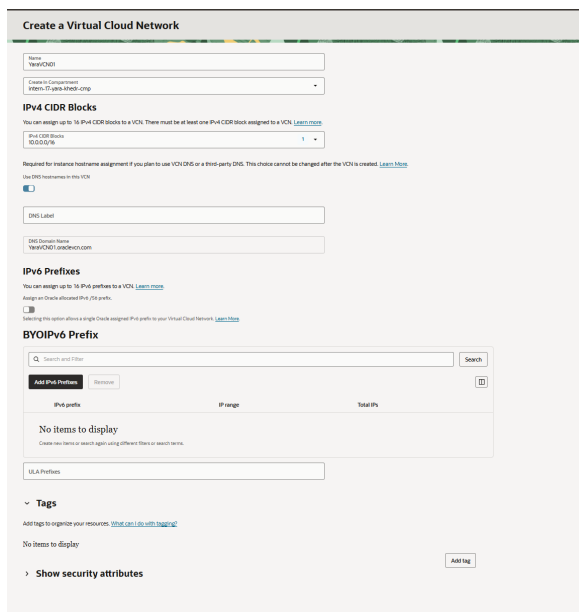
Component	Name	Purpose
VCN	YaraVCN01	Private network space (10.0.0.0/16)
Internet Gateway	YaraIGW01	Enables internet connectivity for the VCN
Route Table	YaraRT-Pub01	Routes 0.0.0.0/0 traffic to the Internet Gateway
Public Subnet	YaraSubnet-Pub01	Hosts all compute and storage resources (10.0.1.0/24)
Security List	YaraSL-NFS	Controls inbound/outbound traffic (SSH + NFS rules)
Compute Instance	YaraVM01	Oracle Linux 8 ARM VM (VM.Standard.A1.Flex)
Block Volume	YaraBlockVolume01	50 GB persistent block storage attached to VM
File System	YaraFS01	OCI File Storage Service (NFS-based shared storage)
Mount Target	YaraMT01	NFS endpoint (Private IP: 10.0.1.154)
Export	YaraExp01	Exposes the file system at path <code>/yara-fs</code>

Table 2.1: Infrastructure Component Summary

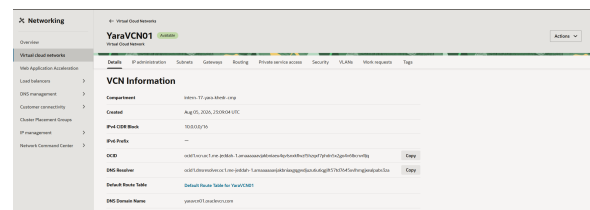
3. Network Setup

3.1 VCN Configuration

A Virtual Cloud Network (VCN) named **YaraVCN01** was created in compartment **intern-17-yara-khedr-cmp** with CIDR block 10.0.0.0/16.



(a) VCN Creation Form

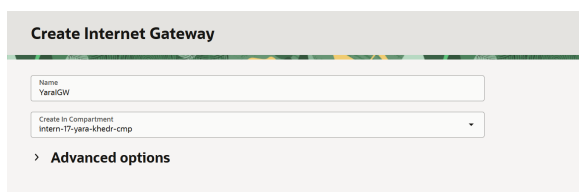


(b) VCN Successfully Created

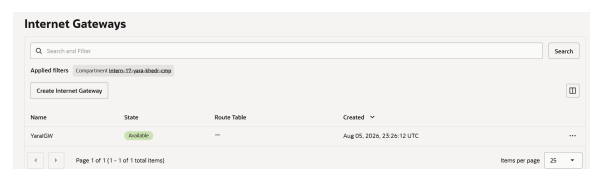
Figure 3.1: VCN Configuration – OCI Console

3.2 Internet Gateway

An Internet Gateway named **YaraIGW01** was attached to the VCN to enable bidirectional internet connectivity. It acts as the single point of entry and exit for all internet-bound traffic within the VCN.



(a) Internet Gateway Creation Form



(b) Internet Gateway Successfully Created

Figure 3.2: Internet Gateway Configuration – OCI Console

3.3 Public Subnet Configuration

A public subnet named **YaraSubnet-Pub01** was created with CIDR block 10.0.1.0/24, providing a smaller network slice within the VCN for hosting compute resources. The subnet was configured as a regional subnet and associated with the route table and security list described below.

(a) Subnet Creation Form

Name	State	IPv4 CIDR Block	IPv6 Prefixes	Subnet Access	Created
YaraSub-Public	Available	10.0.1.0/24	None	Public (Regional)	Aug 07, 2024, 23:55:10 UTC

(b) Subnet Successfully Created

Figure 3.3: Public Subnet Configuration – OCI Console

3.3.1 Route Table

A dedicated route table named **YaraRT-Pub01** was created and associated with the public subnet. It contains a single rule directing all outbound internet traffic (0.0.0.0/0) through the Internet Gateway, making the subnet “public” in the OCI networking model.

(a) Route Table Creation Form

Name	State	Number of Rules	Created
YaraRT-Public	Available	1	Aug 07, 2024, 23:57:06 UTC
Default Route Table for YaraVCN01	Available	0	Aug 07, 2024, 23:00:04 UTC

(b) Route Table Successfully Created

Figure 3.4: Route Table Configuration – OCI Console

3.4 Security List

A custom security list named **YaraSL-NFS** was created and attached to the public subnet to replace the default security list. It defines the following ingress and egress rules:

3.4.1 Ingress Rules

Rule #	Protocol	Source	Port(s)	Purpose
1	TCP	0.0.0.0/0	22	SSH access from internet to compute instance
2	TCP	10.0.1.0/24	111	NFS portmapper/rpcbind (required for NFS handshake)
3	TCP	10.0.1.0/24	2048–2050	NFS data transfer (OCI FSS specific ports)
4	UDP	10.0.1.0/24	111	NFS portmapper/rpcbind over UDP
5	UDP	10.0.1.0/24	2048	NFS data transfer over UDP

Table 3.1: Security List – Ingress Rules

Create Security List

A security list contains ingress and egress rules that specify the types of traffic allowed in and out of instances. [Learn more about Security Lists.](#)

Name:

Create in Compartment:

Allow Rules for Ingress

Ingress Rule 1

Allows TCP traffic for ports: all

Stateless ☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. [Learn about stateful and stateless rules.](#)

Source Type: Source CIDR: IP Protocol:

Source Port Range: Destination Port Range:

Description:

Ingress Rule 2

Allows TCP traffic for ports: all

Stateless ☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. [Learn about stateful and stateless rules.](#)

Source Type: Source CIDR: IP Protocol:

Source Port Range: Destination Port Range:

Description:

Ingress Rule 3

Allows UDP traffic for ports: all

Stateless ☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. [Learn about stateful and stateless rules.](#)

Source Type: Source CIDR: IP Protocol:

Source Port Range: Destination Port Range:

Description:

Ingress Rule 4

Allows UDP traffic for ports: all

Stateless ☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. [Learn about stateful and stateless rules.](#)

Source Type: Source CIDR: IP Protocol:

Source Port Range: Destination Port Range:

Examples: 80, 20-22
View IANA list of [ports](#).

Description:

Figure 3.5: Security List – Ingress Rules (OCI Console)

3.4.2 Egress Rules

OCI Console (Lab 1): Egress rules were defined explicitly for each NFS-related protocol and port, mirroring the granularity of the ingress rules. The full egress rule set is shown in the screenshot below.

Terraform (Lab 2): A single catch-all egress rule was defined (`protocol = "all"`, destination `0.0.0.0/0`), which allows all outbound traffic.

Rule #	Protocol	Destination	Port(s)	Purpose
1	TCP	10.0.1.0/24	111, 2048–2050	NFS outbound portmapper and data transfer
2	UDP	10.0.1.0/24	111, 2048	NFS outbound portmapper and data transfer
3	All	0.0.0.0/0	All	Allow all remaining outbound traffic (Terraform: single rule)

Table 3.2: Security List – Egress Rules (Console: explicit; Terraform: catch-all)

Allow Rules for Egress

Egress Rule 1

Allows TCP traffic for ports: all

Stateless

☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Destination Type

CIDR

Destination CIDR

10.0.10/24

IP Protocol

TCP

Source Port Range

All

Destination Port Range

!!!

Description

Egress Rule 2

Allows TCP traffic for ports: all

Stateless

☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Destination Type

CIDR

Destination CIDR

10.0.10/24

IP Protocol

TCP

Source Port Range

All

Destination Port Range

2048-2050

Description

Egress Rule 3

Allows UDP traffic for ports: all

Stateless

☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Destination Type

CIDR

Destination CIDR

10.0.10/24

IP Protocol

UDP

Source Port Range

All

Destination Port Range

!!!

Description

Egress Rule 4

Allows UDP traffic for ports: all

Stateless

☐

To enable bidirectional traffic flow, make sure a complementary rule in the opposite direction exists. Learn about [stateful](#) and [stateless](#) rules.

Destination Type

CIDR

Destination CIDR

10.0.10/24

IP Protocol

UDP

Source Port Range

All

Destination Port Range

2048

Description

Figure 3.6: Security List – Egress Rules (OCI Console)

YaraSub-Public

Available

Actions

Subnet

Details

IP administration

Private service access

Security

Monitoring

Tags

Security Lists

Q Search and Filter

Search

Applied filters

Add Security List

11

Name	State	Compartment	Created	
YaraNFSRules	Available	intern-17-yara-khedi-cmp	Aug 06, 2026, 20:59:21 UTC	...
Default Security List for YaraVNO1	Available	intern-17-yara-khedi-cmp	Aug 05, 2026, 23:09:04 UTC	...

<

>

Page 1 of 1 (1 - 2 of 2 total items)

Items per page

25

Figure 3.7: Security List Attached to Public Subnet

4. Compute Instance Deployment

4.1 Instance Details

A Linux compute instance named **YaraVM01** was deployed using the following specifications:

Parameter	Value
Instance Name	YaraVM01
Shape	VM.Standard.A1.Flex (ARM-based Ampere)
OCPUs	1
Memory	6 GB
Operating System	Oracle Linux 8.10 (aarch64 / ARM64)
Image OCID	ocid1.image.oc1.me-jeddah-1.aaaaaaaatcefj2cvfk62gorm...
Subnet	YaraSubnet-Pub01 (10.0.1.0/24)
Public IP	158.101.229.133
Availability Domain	oXVt:ME-JEDDAH-1-AD-1

Table 4.1: Compute Instance Specifications

Create compute instance

Basic information

Security

Networking
Required

Storage
Required

Review

Basic information

Edit

Name

YaraInstance01

Create in compartment

intern-17-yara-khedr-cmp

Placement

Availability domain

AD-1

Capacity type

on-demand

Fault domain

Let Oracle choose the best fault domain

Image

Operating system

Oracle Linux 9

Image build

2026.07.20-0

Security

Shape

Shape

VM.Standard.A1.Flex

Shape build

Virtual machine, 1 core OCPU, 6 GB memory, 1 Gbps network bandwidth

Management

Authorization header

Enabled

Tags

Namespace	Key	Value
No items to display		

Figure 4.1: Instance Configuration – Shape and Image Selection

Availability configuration

Live migration	Let Oracle choose the best migration option
Restart after infrastructure maintenance	Enabled

Oracle Cloud Agent

WebLogic Management Service	Disabled
Vulnerability Scanning	Disabled
Oracle Java Management Service	Disabled
OS Management Hub Agent	Disabled
Management Agent	Disabled
Fleet Application Management Service	Disabled
Custom Logs Monitoring	Enabled
Compute RDMA GPU Monitoring	Disabled
Compute Instance Run Command	Enabled
Compute Instance Monitoring	Enabled
Compute HPC RDMA Auto-Configuration	Disabled
Compute HPC RDMA Authentication	Disabled
Cloud Guard Workload Protection	Enabled
Block Volume Management	Disabled
Bastion	Disabled
Secure Boot	Disabled
Measured Boot	Disabled
Trusted Platform Module	Disabled

Figure 4.2: Instance Configuration – Network and Subnet Placement

Security attributes

Namespace	Key	Value
No items to display Create new items or search again using different filters or search terms.		

Networking

Edit

VNIC name	-
Virtual cloud network	YaraVCN01
Subnet	YaraSub-Public
Launch options	-
DNS record	Yes
Use network security groups to control traffic	No
Public IPv4 address	Yes
Private IPv4 address	Automatically assigned on creation
Private IPv4 OCid	-
IPv6 address	undefined
IPv6 address OCID	—

SSH keys

SSH keys	ssh-rsa AA...
----------	---------------

Storage

Edit

Boot volume

Specify a custom boot volume size and performance setting	-
Boot volume performance (VPU)	-
Use in-transit encryption	Enabled
Master encryption key	-

Block volume

Name	Attachment type	Create type	Reservations	Access
No items to display Create new items or search again using different filters or search terms.				

Figure 4.3: Instance Configuration – SSH Key and Additional Options

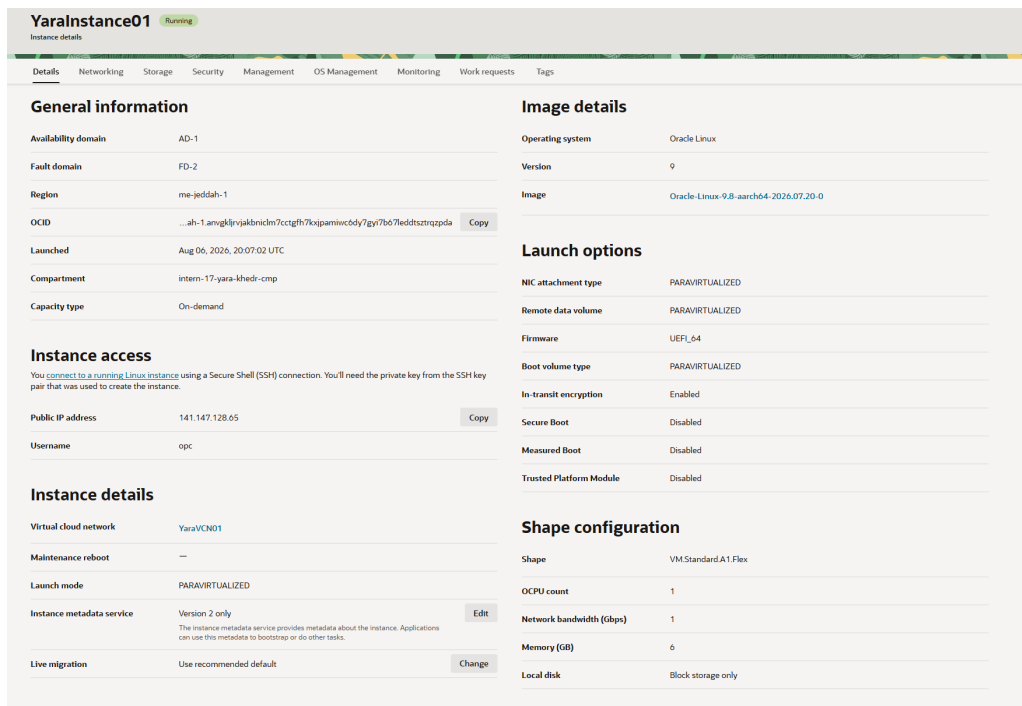


Figure 4.4: Compute Instance Successfully Created and Running

4.2 SSH Key Setup

Access to the Oracle Linux instance is configured via RSA SSH key pair authentication.

Lab 1 (OCI Console): The SSH key pair was generated directly through the OCI Console during instance creation. OCI generated the key pair automatically and provided the private key as a download.

Lab 2 (Terraform): The SSH key pair was generated manually via the terminal using the following command:

```
ssh-keygen -t rsa -b 2048 -f C:\Users\yaraa\.ssh\lab2_key
```

The **public key** was hardcoded into the Terraform `compute.tf` file under the `metadata` block and passed to OCI during instance creation. The **private key** remains on the local machine and is never shared or committed to version control.

4.3 Connecting to the Instance

The instance is accessed via SSH using the `opc` user and the private/public key (depending on network):

```
ssh -i ~/.ssh/lab2_key opc@158.101.229.133
```

Upon successful connection, the terminal prompt confirms access to the running instance:

```
[opc@YaraVM01 ~]$
```

5. File Storage Service (NFS) Setup

5.1 File System and Mount Target Creation

The OCI File Storage Service requires three separate resources:

- **File System** – the actual storage resource
- **Mount Target** – the NFS network endpoint within the subnet
- **Export** – maps the file system to a mountable path through the mount target

A file system named **YaraFS01** and a mount target named **YaraMT01** were created within the public subnet.

The screenshot shows the 'Create file system' form in the OCI Console. It includes sections for 'File system information' (Name: YaraFS01, Compartment: orcm-17-pub-kh01-cmp, Availability domain: ovmcme-jedgah-lad-1), 'Encryption' (options for Oracle-managed or customer-managed keys), 'Attach snapshot policy' (checkbox), 'Tags' (expandable section), 'Resource locks' (No lock, Delete, or Full), and 'Export information' (Export path: /YaraFS01-export01). It also has checkboxes for 'Use secure export options' and 'Use LDAP for group list'.

(a) File System Creation Form

A large rectangular box containing the text '17_FS_MT_Create_Form.png', indicating the location of the Mount Target Creation Form image.

(b) Mount Target Creation Form

Figure 5.1: File Storage Service Configuration – OCI Console

File systems
YaraFileSystem01
Active
File system

Replication is not enabled for this file system.
This file system does not have an active replication in place. Replication provides protection from regional outages, aids in disaster recovery efforts, and addresses data redundancy compliance requirements.
Learn more.

Details
Exports
Snapshots
Replications
Clones
User and group quota rules
Monitoring
Tags

General information

OCID
...edda_1.aasasasaph5h6xrpjvllq0xvotbmuvwuzdmqvqjrfvqwljr
Copy

Created
Thu, Aug 6, 2026, 20:13:00 UTC

Utilization
0 B

Availability domain
oVtME-JEDDAH-1-AD-1

Compartment
intern-17-yara-khedr-cmp

Clones

Hydration
Complete

Source snapshot
--

Parent file system
--

Clone root
False

Descendants
False

Clone attached status
Detached

Clone count
0

Encryption

Encryption key
Oracle-managed key

Encryption key OCID
--

Encryption key state
--

Replication

Targetable
False

Replication target
--

Source replication status
Disabled

Snapshot policy

The snapshot policy associated with this file system. snapshot policy governs policy based snapshots of associated file systems. It contains a list of snapshot schedules where users can define the frequency of snapshot creation for the associated file systems, and the retention duration of the taken snapshots.

Snapshot policy
--

Snapshot policy state
--

Resource lock

Resource locks are used to prevent certain APIs from being called for the resource. A full lock prevents both updating the resource and deleting the resource. A delete lock prevents deleting the resource.
Learn more

Lock type 1
--

Lock type 2
--

Quota

Quotas control the amounts of storage space that can be consumed by a specific user, by a group or at overall file system level.
Learn more

Quota enforcement state
Disabled

Compartment quota

A compartment-level quota limits the total file storage capacity that can be used within a compartment and its child compartments.
Learn more

Compartment quota enforcement state
Not blocked

File system quota rule

Quota rule that can restrict the logical space utilization at the file system level.
Learn more

Hard quota rule limit
--

Soft quota rule limit
--

Default user and group quota rules

Sets the default limits that a user or group can consume in a file system. This can be overridden by applying specific limits for a user or group.
Learn more

Default group hard quota rule limit
--

Default group soft quota rule limit
--

Default user hard quota rule limit
--

Default user soft quota rule limit
--

Figure 5.2: File System and Mount Target – Successfully Created

5.2 Mounting on the Instance

After connecting to the instance via SSH, the file system was mounted using the following commands:

```
# Create the local mount point directory
sudo mkdir -p /mnt/yara-fs

# Mount the NFS file system using the mount target's private IP
sudo mount 10.0.1.154:/yara-fs /mnt/yara-fs
```

The mount was then verified using the `df -h` command, which confirms the file system is accessible and reports OCI FSS's elastic capacity:

```
[opc@YaraVM01 ~]$ df -h | grep yara
10.0.1.154:/yara-fs 8.0E 0 8.0E 0% /mnt/yara-fs
```

```

Cloud Shell

yaraadelkh@cloudshell:~ (me-jeddah-1)$ chmod 400 "ssh-key-2026-08-06 (2).key"
yaraadelkh@cloudshell:~ (me-jeddah-1)$ ssh -i "ssh-key-2026-08-06 (2).key" opc@141.147.128.65
ssh: connect to host 141.147.128.65 port 22: Connection timed out
yaraadelkh@cloudshell:~ (me-jeddah-1)$ ssh -i "ssh-key-2026-08-06 (2).key" opc@10.0.1.218
The authenticity of host '10.0.1.218 (10.0.1.218)' can't be established.
ECDSA key fingerprint is SHA256:E+esLFuOFrTt0hBr84fBPGK4FLEKmMEYw6e/hxUJlRk.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.218' (ECDSA) to the list of known hosts.
[opc@yarainstance01 ~]$ rpm -qa nfs-utils
nfs-utils-2.5.4-42.0.1.el9.8.aarch64
[opc@yarainstance01 ~]$ sudo mkdir -p /mnt/YaraFileSystem01-export01
[opc@yarainstance01 ~]$ sudo mount yaranfs01.yarasubpublic.yaravcn01.oraclevcn.com:/YaraFileSystem01-export01 /mnt/YaraFileSystem01-export01
Created symlink /run/systemd/system/remote-fs.target.wants/rpc-statd.service → /usr/lib/systemd/system/rpc-statd.service.
[opc@yarainstance01 ~]$ df -h
Filesystem
devtmpfs
tmpfs
tmpfs
efivarfs
/dev/mapper/ocivolume-root
/dev/sda2
/dev/mapper/ocivolume-ole
/dev/sda1
tmpfs
tmpfs
yaranfs01.yarasubpublic.yaravcn01.oraclevcn.com:/YaraFileSystem01-export01
[opc@yarainstance01 ~]$ sudo touch /mnt/YaraFileSystem01-export01/testfile.txt
[opc@yarainstance01 ~]$ ls -l /mnt/YaraFileSystem01-export01
total 0
-rw-r--r--. 1 root root 0 Aug  6 20:55 testfile.txt
[opc@yarainstance01 ~]$

```

Figure 5.3: NFS File System Successfully Mounted – OCI Console Verification

6. Block Volume Setup

6.1 Block Volume Creation

A block volume named **YaraBlockVolume01** was created with a size of **50 GB**.

The screenshot shows the 'Create block volume' form. The 'Name' field is 'YaraBlockVolume01'. The 'Create in compartment' dropdown is 'YaraVM01-Comp'. The 'Availability domain' dropdown is 'AD1-ME-SECURE-1-AD-1'. There is a 'Tags' section with a link to 'Add tags to organize your resources'. The 'Volume size and performance' section has 'Volume size (in GB)' set to '50'. The 'Target volume performance' section has 'Performance based auto-tune' checked. Below this, there are sections for 'Balanced' and 'Performance based auto-tune' with checkboxes and descriptions. At the bottom, there is a table for 'VPU', 'IOPS', and 'Throughput'.

VPU	IOPS	Throughput
10 VPU/GB	3,000 IOPS (60 IOPS/GB)	24 MB/s (480 KB/s/GB)

(a) Block Volume Creation Form

The screenshot shows the 'YaraBlockVolume01' details page. It includes sections for 'Block volume information', 'Auto-tuned performance policies', 'Cross region replication', 'Performance', 'Scheduled backups', and 'Cross region copy target'. The 'Block volume information' section shows details like 'Availability domain', 'Cluster placement group', 'Compartment', 'OCID', 'Created', 'Size', 'Hybrid', 'Encryption key', 'Volume group', 'Reservations', and 'Snapshots'. The 'Auto-tuned performance policies' section shows 'Performance based auto-tune' and 'Detached volume auto-tune'. The 'Cross region replication' section shows 'Managed by' and 'Cross region replication'. The 'Performance' section shows 'Auto-tuned performance' and 'Auto-tuned throughput'. The 'Scheduled backups' section shows 'Managed by' and 'Backup policy'. The 'Cross region copy target' section shows 'Cross region copy target'.

(b) Block Volume Successfully Created

Figure 6.1: Block Volume Configuration – OCI Console

6.2 Attaching via iSCSI

The block volume was attached to **YaraVM01** using the **iSCSI** attachment type.

Attach to instance

Attachment type: **iSCSI**
Use iSCSI storage protocol to attach the volume to the instance

Access type: **Read/write**
Configures the volume attachment as Read/Write, not shareable with other instances. This enables attachment to a single instance only and is the default configuration.

Instance: **Select instance**
Choose instance: **interm-f7-yara-khedr-cmp**
YaraInstance01

OCID: **...ztrqzpd**

Image: **Oracle-Linux-9.8-aarch64-2026.07.20-0**

Shape: **VM.Standard.A1.Flex**

Require chap credentials: ☐

Use iSCSI security protocol chap for authentication between the instance and volume.

Use oracle cloud agent to automatically connect to iSCSI-attached volumes: ☐

Device path: **/dev/oracleoci/oracleadb**

Figure 6.2: Block Volume iSCSI Attachment Form – OCI Console

iSCSI IPv4 commands & information

Use these commands when the virtual network for your primary VNIC is set up with IPv4.

Attach commands:

```
1 sudo iSCSIadm -m node -o new -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -p 169.254.2.2:3260
2 sudo iSCSIadm -m node -o update -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -n node.startup -v automatic
3 sudo iSCSIadm -m node -o update -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -p 169.254.2.2:3260 -l
```

Detach commands:

```
1 sudo iSCSIadm -m node -o delete -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -p 169.254.2.2:3260 -u
```

Cloud Shell session:

```
yara@b10g:cloudshell~$ (m-jeddah) [3] chad 480 ssh-key 2026-08-06-key
yara@b10g:cloudshell~$ (m-jeddah) [3] ssh -i ssh-key-2026-08-06-key ec2@b1.39
the authenticity of host 'b1.39 (10.0.1.39)' can't be established.
ECDSA key fingerprint is SHA256:0bqWkF30p4s0t0p0d/c0w0wQ00000.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'b1.39' (ECDSA) to the list of known hosts.
[yara@b10g:cloudshell~$] sudo iSCSIadm -m node -o new -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -p 169.254.2.2:3260
New iSCSI node [http://b1.39:3260/169.254.2.2:3260] added
[yara@b10g:cloudshell~$] sudo iSCSIadm -m node -o update -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -n node.startup -v automatic
[yara@b10g:cloudshell~$] sudo iSCSIadm -m node -o update -T iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59 -p 169.254.2.2:3260 -l
Login to [iface: default, target: iqn.2015-12.com.oracle:ias:f2569488-ffd3-4449-81eb-a620d6247c59, portal: 169.254.2.2:3260] successful.
[yara@b10g:cloudshell~$]
```

Figure 6.3: iSCSI Connect Commands Provided by OCI Console

6.3 Formatting and Mounting

After running the iSCSI connect commands inside the instance, the new disk appeared as a block device. It was formatted with the XFS filesystem and mounted at `/mnt/YaraBlockVolume01`:

```
[opc@yarainstance01 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0  46.6G  0 disk
├─sda1       8:1      0   100M  0 part /boot/efi
├─sda2       8:2      0    2G   0 part /boot
├─sda3       8:3      0   44.5G  0 part
├─ocivolume-root 252:0    0   29.5G  0 lvm /
└─ocivolume-ole 252:1    0    15G  0 lvm /var/ole

[opc@yarainstance01 ~]$ sudo iscsiadm -m node -o new -T iqn.2015-12.com.oracleiaas:aa465db7-f652-4c70-b046-848509496917 -p 169.254.2.2:3260
sudo iscsiadm -m node -o update -T iqn.2015-12.com.oracleiaas:aa465db7-f652-4c70-b046-848509496917 -n node.startup -v automatic
sudo iscsiadm -m node -T iqn.2015-12.com.oracleiaas:aa465db7-f652-4c70-b046-848509496917 -p 169.254.2.2:3260 -l
New iSCSI node [tcp:[hw=,ip=,net_if=,iscsi_if=default] 169.254.2.2,3260,-1 iqn.2015-12.com.oracleiaas:aa465db7-f652-4c70-b046-848509496917] added
Login to [iface: default, target: iqn.2015-12.com.oracleiaas:aa465db7-f652-4c70-b046-848509496917, portal: 169.254.2.2,3260] successful.
[opc@yarainstance01 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0  46.6G  0 disk
├─sda1       8:1      0   100M  0 part /boot/efi
├─sda2       8:2      0    2G   0 part /boot
├─sda3       8:3      0   44.5G  0 part
├─ocivolume-root 252:0    0   29.5G  0 lvm /
└─ocivolume-ole 252:1    0    15G  0 lvm /var/ole
sdb          8:16     0   50G   0 disk
[opc@yarainstance01 ~]$ sudo mkfs.xfs /dev/sdb
meta-data=/dev/sdb          isize=512    agcount=4, agsize=3276800 blks
               =                sectsz=4096   attr=2, projid32bit=1
               =                crc=1        finobt=1, sparse=1, rmapbt=0
               =                reflink=1     bigtime=1 inobtcount=1 nrext64=0
               =                exchange=0    metadir=0
data        =                bsize=4096     blocks=13107200, imaxpct=25
               =                sunit=0      swidth=0 blks
naming      -version 2        bsize=4096   ascii-ci=0, ftype=1, parent=0
log         -internal log    bsize=4096   blocks=16384, version=2
               =                sectsz=4096   sunit=1 blks, lazy-count=1
realtime    =none           extsz=4096   blocks=0, rtextents=0
               =                rgcount=0     rgsz=0 extents
               =                zoned=0       start=0 reserved=0
Discarding blocks...Done.
[opc@yarainstance01 ~]$ sudo mkdir -p /mnt/YaraBlockVolume01
[opc@yarainstance01 ~]$ sudo mount /dev/sdb /mnt/YaraBlockVolume01
[opc@yarainstance01 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0      0  46.6G  0 disk
├─sda1       8:1      0   100M  0 part /boot/efi
├─sda2       8:2      0    2G   0 part /boot
├─sda3       8:3      0   44.5G  0 part
├─ocivolume-root 252:0    0   29.5G  0 lvm /
└─ocivolume-ole 252:1    0    15G  0 lvm /var/ole
sdb          8:16     0   50G   0 disk /mnt/YaraBlockVolume01
[opc@yarainstance01 ~]$ df -h /mnt/YaraBlockVolume01
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb         50G   389M   50G   1% /mnt/YaraBlockVolume01
[opc@yarainstance01 ~]$
```

Figure 6.4: Block Volume Mounted and Verified – lsblk and df -h Output

7. Terraform Automation (Lab 2)

7.1 Overview

Lab 2 reproduces the entire environment built manually in Lab 1 using Terraform Infrastructure as Code. The complete Terraform configuration is version-controlled and publicly available at:

<https://github.com/Yara-Khedr/ejada-intern-w1-terraform>

7.2 Repository Structure

The Terraform project is organized into the following files, each with a single clear responsibility:

File	Purpose
<code>provider.tf</code>	Declares the OCI Terraform provider (oracle/oci) and configures API Key authentication using the local <code>~/.oci/config</code> file
<code>variables.tf</code>	Declares all input variables (compartment OCID) with descriptions and types, keeping values separate from resource definitions
<code>network.tf</code>	Defines all networking resources: VCN, Internet Gateway, Route Table, Public Subnet, and Security List with NFS and SSH rules
<code>compute.tf</code>	Defines the Oracle Linux ARM compute instance with shape config, source image, VNIC details, and SSH key metadata
<code>storage.tf</code>	Defines all storage resources: Block Volume, paravirtualized Volume Attachment, FSS File System, Mount Target, and Export
<code>outputs.tf</code>	Declares Terraform outputs printed after every apply – the instance public IP and mount target private IP
<code>.gitignore</code>	Excludes sensitive and generated files from Git: <code>.terraform/</code> , <code>*.tfstate</code> , <code>*.tfvars</code> , <code>*.pem</code>
<code>.terraform.lock.hcl</code>	Records exact provider versions to ensure reproducible <code>terraform init</code> runs across environments

Table 7.1: Terraform Repository File Structure

7.3 Terraform Workflow

The standard Terraform CLI workflow was used to deploy the environment:


```
terraform init # Download OCI provider plugin (oracle/oci v8.26.0)
terraform plan # Preview changes before applying
terraform apply # Create all resources in OCI
terraform destroy # Tear down all managed resources
```

Each resource was built and verified incrementally, running `terraform plan` before every `apply` to confirm expected changes.

7.4 Terraform Outputs

After a successful `terraform apply`, the following outputs were printed:

Outputs:

```
MT_Priv_IP = "10.0.1.154"
VM_Pub_IP = "158.101.229.133"
```

7.5 File Storage NFS Mount via Terraform Environment

The file system provisioned by Terraform was successfully mounted on the compute instance, confirming end-to-end infrastructure functionality:

```
[opc@YaraVM01 ~]$ sudo mkdir -p /mnt/yara-fs
[opc@YaraVM01 ~]$ sudo mount 10.0.1.154:/yara-fs /mnt/yara-fs
mount.nfs: Connection timed out
[opc@YaraVM01 ~]$ sudo mount 10.0.1.154:/yara-fs /mnt/yara-fs
[opc@YaraVM01 ~]$ df -h | grep yara
10.0.1.154:/yara-fs      8.0E    0  8.0E    0% /mnt/yara-fs
[opc@YaraVM01 ~]$ |
```

Figure 7.1: NFS File System Successfully Mounted on Terraform-Provisioned Instance